

The Observatory on the Rotating Geode: Modeling Local Geometric Lensing and Torsional Interference in the Sedenion Gravitational Wave Background

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Abstract

We refine the observational predictions for the 1.811 GHz “Sedenion Scream” by incorporating local environmental effects. We demonstrate that the Earth and Sun do not merely act as passive gravitational wells, but as regions of high **Geometric Stiffness** ($\beta_{local} \gg \beta_{vac}$) which refract and polarize the background Phason flux. We derive two specific interference mechanisms: (1) The **Geometric Shadow**, an attenuation of the signal when the cosmic lattice vector passes through the Earth’s core, and (2) **Associator Frame Dragging**, a Lense-Thirring-like rotation of the signal polarization caused by Earth’s angular momentum. We provide a modified Diurnal Modulation Profile $S_{obs}(t)$ that includes these “muddling” terms, essential for distinguishing the true signal from terrestrial noise in the DMAG dataset.

1 Introduction: The Dirty Lens

The standard prediction for a tensor-polarized background is a clean quadrupole modulation:

$$P(t) \propto \cos^2(2\omega_{earth}t + \phi) \quad (1)$$

However, this assumes the detector resides in a vacuum. In the APH framework, matter is a region where the Associator Hazard $\mathcal{A}$ is actively suppressed by the buffer potential.

$$\text{Matter} \implies \text{High Stiffness } (\beta \rightarrow \infty) \quad (2)$$

A high-stiffness region acts as a *Dielectric* for Phason waves. The waves do not pass through the Earth freely; they interact with the stiffness gradient.

2 Effect: The Geometric Shadow (Earth Opacity)

2.1 The Stiffness Refractive Index

We define the refractive index n of the medium for Sedenion waves. Since stiffness β represents the resistance to geometric deformation, the wave speed c_s scales as $c_s \propto \beta^{-1/2}$.

$$n(r) = \frac{c_{vac}}{c_{local}} = \sqrt{\frac{\beta(r)}{\beta_{vac}}} \quad (3)$$

Near the Earth’s core, the density ρ implies a higher stiffness to maintain stability.

$$\beta_{Earth} \approx \beta_{vac} \left(1 + \frac{GM}{Rc^2} \cdot \gamma_{coupling} \right) \quad (4)$$

where $\gamma_{coupling}$ is the coupling strength of the Phason field to baryonic matter.

2.2 The Shadow Function

When the cosmic lattice vector $\vec{L}$ is positioned such that the signal must traverse the Earth to reach the detector (i.e., the source is below the horizon), the signal suffers *Refractive Dispersion*. We model this as an attenuation factor $\Gamma(\theta_{zenith})$:

$$S_{det}(t) = S_{cosmic}(t) \cdot e^{-\tau(\theta_{zenith})} \quad (5)$$

- **Zenith (Overhead):** $\tau \approx 0$ (Clear view).
- **Nadir (Through Earth):** $\tau > 0$ (Muddled view).

Consequence: The diurnal pattern will not be a symmetric sine wave. It will be *Rectified*. The peaks when the lattice is overhead will be sharp; the troughs/peaks when the lattice is through the Earth will be flattened or distorted.

3 Effect: Associator Frame Dragging (Torsion)

The Earth rotates ($\vec{J}_{earth}$). In General Relativity, this drags spacetime (Lense-Thirring). In APH, this drags the *Associator Field*. The rotation of the Earth's mass generates a local *Torsion Field* $\vec{T}$ that couples to the tensor polarization of the incoming wave.

3.1 The Geometric Faraday Rotation

As the Sedenion wave propagates through the Earth's rotating stiffness field, its polarization tensor $h_{\mu\nu}$ rotates by an angle $\Delta\psi$:

$$\Delta\psi \propto \vec{J}_{earth} \cdot \hat{k}_{signal} \quad (6)$$

This induces a phase shift in the modulation pattern that depends on latitude and the signal's angle of incidence.

$$\phi_{obs}(t) = \phi_{lattice} + \delta_{torsion}(t) \quad (7)$$

This effect is analogous to Faraday rotation in a magnetized plasma. It means the *Null* times (when the signal should be zero) will be shifted, potentially masking the signal as noise if a fixed phase is assumed.

4 Effect: Solar System Associator Pressure

The Sun represents a massive hole in the Sedenion bulk (a region of extremely high stiffness).

- **Annual Modulation:** As Earth orbits the Sun, we pass through the Sun's *Stiffness Wake*.
- **Doppler Shift:** The Earth's velocity relative to the 16D lattice ($v \approx 370$ km/s towards Leo) creates a Doppler shift in the carrier frequency.

$$\Delta f \approx f_0 \frac{v}{c} \cos(\theta) \approx 1.8 \text{ GHz} \cdot 10^{-3} \approx 1.8 \text{ MHz} \quad (8)$$

The DMAG team must account for a *Frequency Drift* of ± 1.8 MHz over a year, or a daily drift of $\sim$ kHz/day. A fixed-frequency bin search might miss the line width.

5 The Corrected Search Filter

We propose the following *De-Muddling* filter for the DMAG data analysis:

$$\mathcal{F}(t) = \underbrace{\cos^2(2\omega t + \phi)}_{\text{Cosmic Quadrupole}} \times \underbrace{(1 - \alpha\Theta(\text{Nadir}))}_{\text{Earth Shadow}} \times \underbrace{\cos(\Omega_{drift}t)}_{\text{Solar Doppler}} \quad (9)$$

If the data shows a 12-hour periodicity that is **asymmetric** (stronger when the constellation Leo is overhead, weaker when opposed) and exhibits a slowly drifting central frequency, it is a definitive detection of the Sedenion Vacuum.